Date - 30-10-26

1480 : Running Sum of 1D Array.

Input: nums = [1,2,3,4] , Output: [1,3,6,10]

using loop to iterate each element of array through index

res[0] = nums[0]

res[i] = res[i-1] + nums[i]

for loop i = 1 to length - 1

i = 1 res[1] = res[0] + nums[1]
= 1+2
= 3

i = 2 res[2] = res[1] + nums[2]
= 3+3
= 6

i = 3 res[3] = res[2] + nums[3]
= 6+4
= 10

return result

Time Complexity: O(n)

Code

```
Class Solution {
    public int[] runningSum(int[] nums){
        int[] res = new int[nums.length];
        res[0] = nums[0];
        for(int i=1; i< nums.length; i++){
            res[i] = res[i-1]+nums[i];
        }
        return result;
    }
}
```

it take extra Space O(n)

Optimized soln in term of time and space

```
Class Solution {
    public int[] runningSum(int[] nums){
        for(int i=1; i< nums.length; i++){
            nums[i] = nums[i] + nums[i-1];
        }
        return result;
```

Space Complexity O(1)

Problem List

Submit

Premium

Description | Accepted | Editorial | Solutions | Submissions

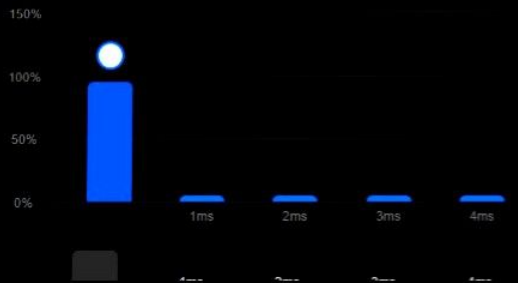
All Submissions

**Runtime**

**0** ms | Beats **100.00%**

Memory

43.93 MB | Beats 80.06%




Code

Java | Auto

```java
class Solution {
    public int[] runningSum(int[] nums) {

        for(int i=1; i<nums.length; i++){
            nums[i] = nums[i] + nums[i-1];
        }
        return nums;
    }
}
```

Saved

Ln 7, Col 9

Testcase | Test Result

**Accepted** Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

nums =

[1,2,3,4]